

Build An IoT-Based Motion Detector Using Cayenne

Presented here is a motion-sensor-based security system using Cayenne and Arduino Uno. Cayenne is the simplest and easiest platform for the Internet of Things (IoT) systems. It allows you to connect regular boards like Raspberry Pi, Arduino and LoRa, and does not require much coding except some editing in readily-available codes. The code (sketch) is automatically generated by Cayenne and, hence, there is no additional code provided for this project.

Circuit and working

The circuit diagram of the motion-sensor-based security alert system is shown in Fig. 1. It is built around Arduino Uno board (Board1), 5V single-changeover relay, PIR motion sensor and a few other components.

Assemble the components on a breadboard or veroboard as per the circuit diagram. The author's prototype with relay circuit wired on a veroboard is shown in Fig. 2.

Cayenne setup and testing

1. Open <https://developers.mydevices.com/cayenne/signup/> and create an account using a valid email address.

2. Choose a device to start. In this case, make sure you select Arduino as the device (Fig. 3).

3. Add Cayenne library to Arduino from Arduino setup option.

4. After selecting Arduino, you will see a list of Arduino boards; select Arduino Uno→Serial USB Connection→Sketch (Fig. 4). The code will be generated automatically. Name the code; in this case it is Serial_USB_Code.ino. Copy and paste this code to Arduino IDE.

5. Connect Arduino Uno board to the PC/laptop using USB cable, and select proper board and COM port under Tools menu in Arduino IDE.

6. Upload Serial_USB_Code.ino code to Arduino board.

7. For USB serial connection, run a script in command prompt so that the board is connected to Cayenne via USB and PC. This provides an Internet connection to the board, which is already present on the Windows PC.

To run scripts, go to Arduino→Libraries→Cayenne→Extras→Scripts folder and note the address from the address bar.

Then, open command prompt and use this address to navigate the

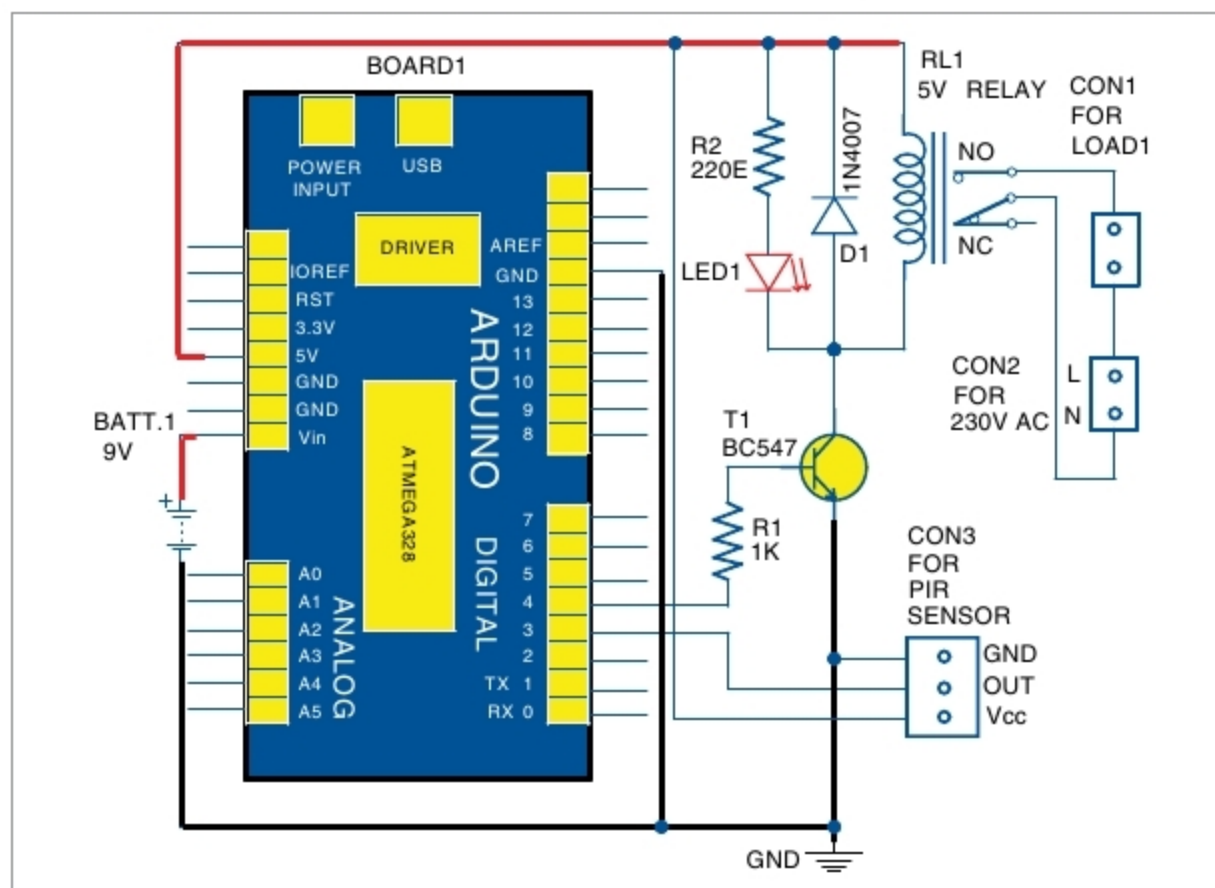


Fig. 1: Circuit diagram of the motion-sensor-based security alert system

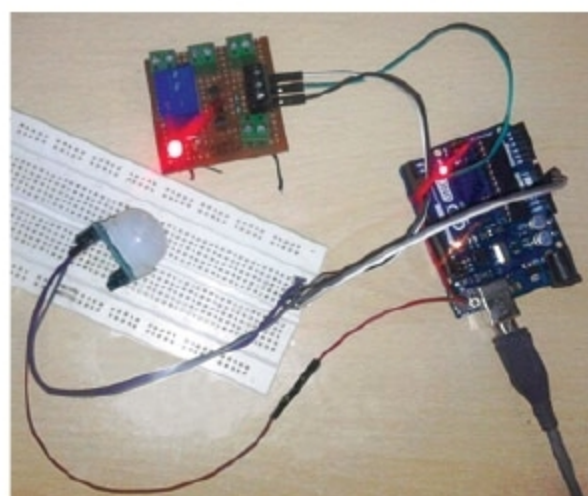


Fig. 2: Author's prototype

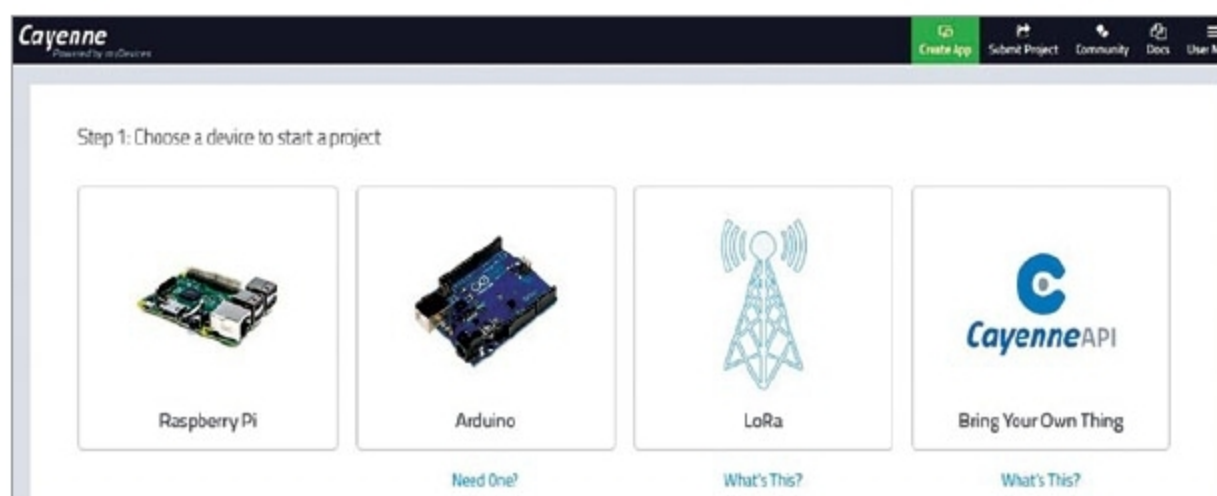


Fig. 3: Device selection on Cayenne